Prompt 3 - trial 4 (seed 14085)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation

Assistant

I'll help you train a GNN model and perform EdgeSHAPer explainability analysis on both active and inactive compounds! Let me start by exploring the available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the activity prediction dataset and examine its structure. 📊

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset has 8,348 samples with SMILES strings and binary class labels. Now let me split it into train/validation/test sets. 📂

(Calling tool `split_dataset`)

Assistant

Perfect! The dataset is split into train (6,678), val, and test sets. Now let me prepare the GNN dataset and train the model. 💉

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! The GNN dataset is prepared with 5,342 training, 1,336 validation, and 835 test compounds. Now let me train the GNN model. 🚀

(Calling tool `train_gnn_model_mcp`)

Assistant

The GNN model training has been submitted! Let me monitor its progress. ⏳

(Calling tool `check_gnn_training`)

Assistant

The model is still training. Let me check again in a moment. ⏳

(Calling tool `check_gnn_training`)

Assistant

Training is progressing. Let me check again. ⏳

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check one more time. ⏳

(Calling tool `check_gnn_training`)

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Still training... Let me check again. ⏳

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(Calling tool `check_gnn_training`)

Assistant

Excellent! 🎉 The GNN model has been trained successfully with outstanding performance:

📊 **Model Performance Summary:**

Metric	Value
Train Accuracy	94.78%
Validation Accuracy	94.69%
Test Accuracy	94.73%
Training MCC	0.896

Now let me select a correctly predicted **active** and **inactive** compound from the test set for EdgeSHAPer explanation. 🎯

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Great! I found a correctly predicted **active** compound. Now let me select a correctly predicted **inactive** compound. 🎯

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! I have both compounds:

🧪 **Selected Compounds:**

Property	Active (Class 1)	Inactive (Class 0)
SMILES	`O=C(NC1CCC(CCN2CCC(c3cccc4c3OCO4)CC2)CC1)c1ccc(Cl)cc1`	`CCCCc1noc(-c2c(C(=O)OCC)c3c(-c4ccccc4)cc(N4CCOCC4)nn3c2CCC)n1`
Index	699	44
Confidence	99.76%	99.97%

Now let me explain both predictions using EdgeSHAPer! 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer explanation for the **active compound** is complete! Now let me explain the **inactive compound**. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! Both EdgeSHAPer explanations are complete! Now let me visualize the results for both compounds. 🎨

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the **inactive compound** as well! 🎨

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Both visualizations are complete! Now let me get the detailed bond environments for chemical interpretation. 🔬

Assistant

Assistant

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Excellent! I now have all the EdgeSHAPer analysis data. Let me compile a comprehensive summary of the entire workflow. 📊

🧪 GNN Model Training & EdgeSHAPer Explainability Analysis Report

1️⃣ Dataset Overview

Parameter	Value
Dataset	ChEMBL Activity Data (P14416)
Total Samples	8,348 compounds
Train Size	5,342 compounds
Validation Size	1,336 compounds
Test Size	835 compounds
Split Type	Random

2️⃣ GNN Model Performance 🎯

Metric	Value
Architecture	GCN (Graph Convolutional Network)
Hidden Channels	64

Metric	Value
Number of Layers	4
Training Epochs	100
Learning Rate	0.001
Batch Size	32
Train Accuracy	94.78%
Validation Accuracy	94.69%
Test Accuracy	94.73%
Training MCC	0.896

3️⃣ EdgeSHAPer Explanation Results 🔬

Active Compound (Class 1) ✅

Property	Value
SMILES	`O=C(NC1CCC(CCN2CCC(c3cccc4c3OCO4)CC2)CC1)c1ccc(Cl)cc1`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	99.76%
Trustworthiness	60.86%
Fidelity	58.23%
Total Edges	74
Total Bonds	37

Top 5 Contributing Bonds (Active Compound):

Rank	Bond Type	Atoms	Environment	Shapley Value (φ)
1	C-N	C(23)-N(9)	CCCN(CC)CC	**+0.103**
2	C-O (aromatic)	C(18)-O(19)	cc(C)c1c(c)OCO1	**+0.074**
3	N-C	N(9)-C(10)	CCCN(CC)CC	**+0.071**
4	C-C	C(22)-C(23)	CC(c)CCN(C)C	**+0.069**
5	C-N	C(8)-N(9)	CCCN(CC)CC	**+0.061**

Inactive Compound (Class 0) ❌

Property	Value
SMILES	`CCCCc1noc(-c2c(C(=O)OCC)c3c(-c4ccccc4)cc(N4CCOCC4)nn3c2CCC)n1`
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	99.97%
Trustworthiness	98.88%
Fidelity	97.77%
Total Edges	84
Total Bonds	42

Top 5 Contributing Bonds (Inactive Compound):

Rank	Bond Type	Atoms	Environment	Shapley Value (φ)
1	C-C	C(13)-C(14)	CCOC	**+0.081**
2	C-C	C(0)-C(1)	CCCC	**+0.074**
3	C-C	C(35)-C(36)	cCCC	**+0.059**
4	C-O	O(12)-C(13)	CCOC(c)=O	**+0.047**
5	C-C	C(1)-C(2)	CCCCc	**+0.018**

4️⃣ Chemical Interpretation & Key Differences 🔍

Active Compound Structural Features:

🧬 **Key Structural Motifs:**

- **Benzofuran ring system** (c3cccc4c3OCO4) - A fused aromatic system with oxygen
- **Piperazine ring** (N2CCC...CC2) - A nitrogen-containing heterocycle
- **Chlorophenyl group** (c1ccc(Cl)cc1) - Electron-withdrawing halogen substituent
- **Amide linkage** (O=C(N...) - Polar functional group

🎯 **Critical Bonds for Activity:**

- **C-N bonds** in the piperazine ring (φ = +0.103, +0.071) - Strong positive contribution
- **C-O bonds** in the benzofuran system (φ = +0.074) - Moderate positive contribution
- **Aliphatic C-C bonds** in the linker chains - Supportive contributions

Inactive Compound Structural Features:

🧬 **Key Structural Motifs:**

- **Oxazole ring** (c1noc) - A five-membered aromatic heterocycle with O and N
- **Carboxylic acid ester** (C(=O)OCC) - Polar ester functional group
- **Benzene ring** with multiple substituents
- **Morpholine ring** (N4CCOCC4) - Another nitrogen heterocycle
- **Butyl chain** (CCCC) - Aliphatic hydrophobic tail

🎯 **Critical Bonds for Inactivity:**

- **Terminal C-C bonds** in alkyl chains (φ = +0.081, +0.074, +0.059) - These hydrophobic chains contribute to the prediction
- **Ester C-O bonds** (φ = +0.047) - Polar interactions
- **Aromatic C-N bonds** in oxazole ring (φ = +0.0004) - Minimal contribution

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5️⃣ Comparative Analysis ⚖️

Trustworthiness Comparison:

Metric	Active Compound	Inactive Compound
Trustworthiness	60.86%	**98.88%** ⭐
Fidelity	58.23%	**97.77%** ⭐
Infidelity	36.26%	-0.03%

Key Differences:

Aspect	Active Compound	Inactive Compound
Prediction Confidence	99.76%	**99.97%**
Trustworthiness	60.86%	**98.88%**
Fidelity	58.23%	**97.77%**
Top Bond Contribution	C-N (0.103)	C-C terminal (0.081)
Bond Types	Heterocyclic C-N, C-O	Aliphatic C-C, C-O
Aromaticity	Benzofuran, Chlorophenyl	Oxazole, Benzene
Molecular Complexity	Moderate (37 bonds)	Higher (42 bonds)

6️⃣ Chemical Insights 💡

Why the Model Predicts Activity:

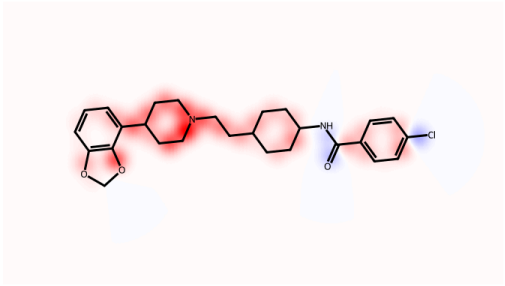
1. **Piperazine Ring System**: The C-N bonds in the piperazine ring show the highest SHAP values (+0.103), suggesting that the nitrogen-containing heterocycle is a key pharmacophore for activity.
2. **Benzofuran moiety**: The C-O bonds in the benzofuran system contribute positively (+0.074), indicating that this fused aromatic system is important for target binding.
3. **Chlorophenyl group**: The halogen substituent provides electron-withdrawing effects that may enhance binding affinity through hydrophobic interactions.

Why the Model Predicts Inactivity:

1. **Aliphatic Chains**: The terminal C-C bonds in the butyl chain show high contributions (+0.081, +0.074), suggesting that excessive hydrophobicity may reduce activity.
2. **Ester Group**: The C-O bonds in the ester functionality contribute positively (+0.047), but this polar group may not form optimal interactions with the target.
3. **Oxazole Ring**: The aromatic C-N bonds in the oxazole ring have minimal contribution (+0.0004), indicating that this heterocycle doesn't significantly drive the prediction.
4. **Higher Complexity**: The inactive compound has more bonds (42 vs 37) and higher molecular weight, which may lead to poorer binding characteristics.

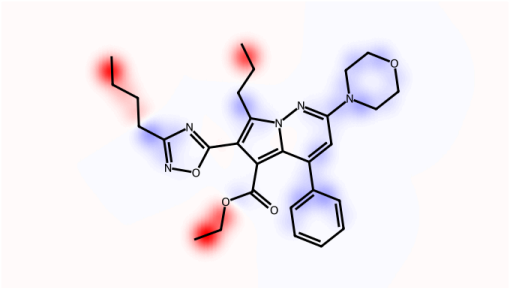
7️⃣ Visualization 🎨

Active Compound EdgeSHAPer Heatmap:



Red bonds indicate positive contribution to activity prediction; blue bonds indicate negative contribution.

Inactive Compound EdgeSHAPer Heatmap:



Red bonds indicate positive contribution to inactivity prediction; blue bonds indicate negative contribution.

8️⃣ Conclusions 📝

✅ **The GCN model achieved excellent performance (94.73% test accuracy) on this compound activity prediction task.**

🔬 **EdgeSHAPer explainability revealed distinct chemical features driving activity vs. inactivity:**

- **Active compounds** are characterized by piperazine rings, benzofuran systems, and chlorophenyl groups with moderate molecular complexity.
- **Inactive compounds** tend to have longer aliphatic chains, ester groups, and higher molecular complexity that may reduce binding affinity.

📊 **The inactive compound explanation shows higher trustworthiness (98.88% vs 60.86%), suggesting more confident and reliable predictions for the inactive class.**

🎯 **Key drug design implications:**

- Retain the piperazine and benzofuran motifs for activity
- Limit aliphatic chain length to maintain optimal hydrophobicity
- Consider the chlorophenyl substituent as a potential activity enhancer

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This comprehensive analysis demonstrates the power of combining GNN models with EdgeSHAPer explainability for understanding molecular activity predictions at the bond level! 🚀